

AA279A FINAL CHEATSHEET

$$\vec{F}_{jk} = -G \frac{m_j m_k}{r_{kj}^2} \hat{r}_{kj} \Rightarrow \ddot{\vec{r}}_j = -G \sum_{\substack{k=1 \\ k \neq j}}^N m_k \frac{\vec{r}_{kj}}{r_{kj}^3} \Rightarrow \ddot{\vec{r}}_2 = G(m_1 + m_2) \frac{\vec{r}_{21}}{r_{21}^3} + \sum_{k=3}^N G m_k \left(\frac{\vec{r}_{2k}}{r_{2k}^3} - \frac{\vec{r}_{1k}}{r_{1k}^3} \right)$$

dominant term perturbation

Newton's Law of Grav. $\Rightarrow$ General N-body formulation $\Rightarrow$ N-body decomposed form.

$\vec{F}_{jk}$: force on j applied by k ; $\vec{r}_{kj}$: source-destination ; -body 1: CB, -body 2: satellite

$$\ddot{\vec{r}} + \frac{M}{r^3} \vec{r} = 0 \quad \text{where} \quad M = G(m_{CB} + m_{sat}) \quad \text{Common } M_s: M_{earth} = 3.986 \times 10^5 \quad M_{sun} = 1.327 \times 10^{20}$$

Restricted 2-body

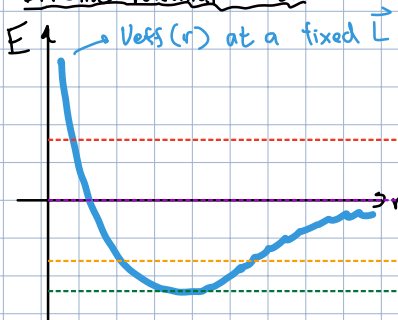
CONSERVED VALUES ($\vec{L}$ & E)

$$\vec{L} = \vec{r} \times m\vec{v} = \vec{r} \times \vec{p} = m\omega r^2$$

$$E = \frac{1}{2} m v^2 + \frac{1}{2} \frac{L^2}{m r^2} - G \frac{M m}{r}$$

True potential $V(r)$
effective potential $V_{eff}(r)$

Effective Potential Plot.



hyperbolic orbit, E too big, over periapsis

Parabolic orbit, $E=0$ (escape velocity)

Elliptical orbit, one periapsis one apoapsis

circular orbit, one fixed orbital radius

impossible orbit, $\frac{1}{2} m v^2$ term cannot contr. neg. energy.

CIRCULAR ORBITS

$$a = \frac{v^2}{r}$$

$$v_{circ}(r) = \sqrt{\frac{GM}{r}} = \sqrt{\frac{M_{CB}}{r}} = \frac{2\pi r}{T_{circ}}$$

$$T_{circ} = 2\pi \sqrt{\frac{r^3}{\mu}}$$

$$n = \sqrt{\frac{M_{CB}}{r^3}} = \frac{2\pi}{T} \quad [\text{rad/s}]$$

* mean motion

CANONICAL UNITS

$$1 \text{ DU}_{CB} = R_{CB} \quad 1 \text{ TU}_{CB} = \sqrt{\frac{R_{CB}^3}{M_{CB}}}$$

$$v_{circ} = 1 \frac{\text{DU}_{CB}}{\text{TU}_{CB}} \quad M_{CB} = 1 \frac{\text{DU}^3 \text{CB}}{\text{TU}_{CB}}$$

* only $M_{CB} = 1$ holds up at any orbit. Beyond $r > R_{CB}$, $v_{circ} \neq 1 \text{ DU}_{CB}/\text{TU}_{CB}$!

SPECIFIC ENERGY

E is conserved for an object in a conservative field

$$\epsilon = \frac{E}{m} = \frac{1}{2} v^2 - \frac{\mu}{r} \quad [\text{J/kg}]$$

$$\text{if } \epsilon = 0, \quad v = \sqrt{\frac{2\mu}{r}} = \sqrt{2} v_{circ} = v_{esc}$$

$$\text{if } \epsilon > 0, \quad v^2 = v_0^2 + v_{esc}^2$$

where v_0 := excess hyperbolic vel.

ORBITAL ELEM'S DERIVATION (start with $\vec{r}, \dot{\vec{r}}$)

- Find $\vec{h}$ (specific angular momentum): $\vec{h} = \vec{L}/m = \vec{r} \times \dot{\vec{r}}$; $\frac{d\vec{h}}{dt} = 0$ * $\vec{h}$ is constant for any 2-B orbit
- Find $\vec{A}$ (eccentricity vector): $\vec{h} \times \dot{\vec{r}} = -\mu \vec{r}/r - \vec{A}$ * $\vec{A}$ always points towards pericenter
- Find ν (true anomaly): $h^2 = \mu r + |\vec{A}| r \cos(\nu)$
- find p & e : $e = \frac{|\vec{A}|}{\mu}$; $p = \frac{h^2}{\mu}$

$$\text{Find } a \text{ (semi-major axis) by vis-viva: } v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right) \Rightarrow a = \left(\frac{2}{r} - \frac{v^2}{\mu} \right)^{-1}$$

$$\text{Substitute } v^2 \text{ into } \epsilon(\nu, r): \epsilon = -\mu/2a \quad \text{* importantly, } a \text{ fully defines orbital energy}$$

$$\text{Determine eccentric anomaly } E: \tan \frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{E}{2}$$

$$\text{Determine \& propagate mean anomaly } M: E - e \sin E = n(t-t_0) = M; \quad M = M_0 + n(t-t_0)$$

KEPLER'S LAWS

- orbits make ellipses
- Equal swept areas

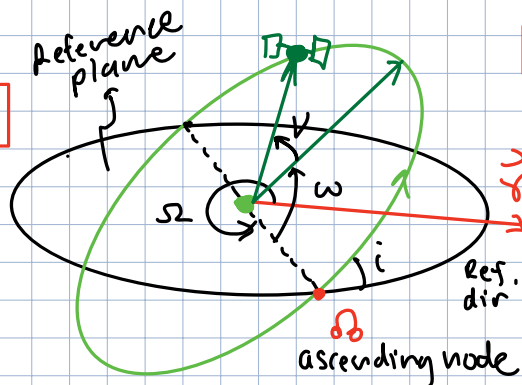
$$\Delta A = \frac{1}{2} |\vec{r} \times \dot{\vec{r}} \Delta T| = \frac{1}{2} \Delta T |\vec{h}|$$

* equal $\Delta T \Rightarrow$ equal ΔA !!

③ T vs a relationship

$$\frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3}; \quad T^2 = 4\pi \frac{a^3}{\mu}$$

OES $\Rightarrow$ QNS OES (start with $[a, e, i, \Omega, \omega, \nu]$)



Aux elems

$$u = \nu + \omega \quad \text{"argument of latitude"}$$

$$\bar{\omega} = \omega + \Omega \quad \text{"longitude of pericenter"}$$

$$l = \bar{\omega} + \nu; \quad L = \bar{\omega} + M$$

QNS OES a - stays same

$$h = e \sin(\bar{\omega}) \quad \left. \begin{array}{l} \text{handles case } e=0, \\ \omega \text{ undef.} \end{array} \right\}$$

$$p = \tan(i/2) \sin \Omega \quad \left. \begin{array}{l} \text{handles case } i=0, \\ \Omega \text{ undefined.} \end{array} \right\}$$

$$l = \bar{\omega} + \nu$$

ALGORITHM: OE 2 Cartesian

① convert from OE $\rightarrow$ XYZ or B.

where $\vec{z}_{\text{orb}} \uparrow \vec{h}$ & $\vec{x}_{\text{orb}} \rightarrow \text{pericenter}$

$$x_{\text{orb}} = r \cos v = \frac{a(1-e^2)}{1+e \cos v} \cos v$$

$$y_{orb} = r \sin \nu = \quad \quad \quad \sin \nu$$

$$\dot{x}_{\text{orb}} = -\frac{\mu}{h} \sin \nu = -\sqrt{\frac{\mu}{a(1-e^2)}} \sin \nu$$

$$\dot{y}_{\text{orb}} = \frac{\mu}{h} (e + \cos v) = \text{" } (e + \cos v)$$

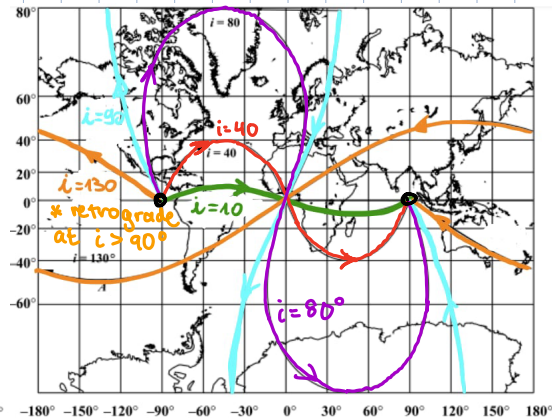
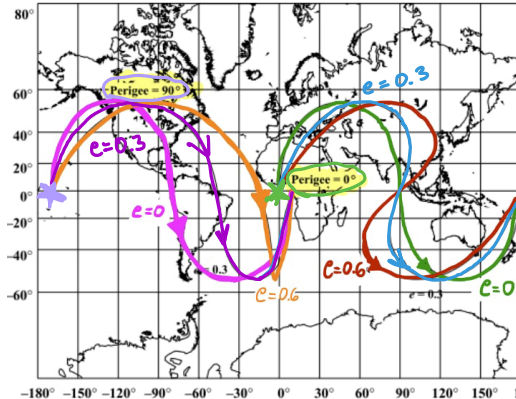
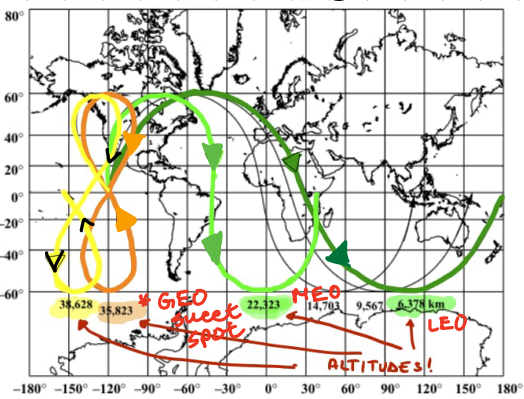
$$z_{orb} = \dot{z}_{orb} = 0$$

② Convert from XYZ ORB to XYZ ECI

$$\begin{bmatrix} \uparrow \\ \downarrow \\ \uparrow \end{bmatrix}_{\text{FCI}} = R_z(-\Omega) R_x(-i) R_z(-\omega) \begin{bmatrix} \uparrow \\ \downarrow \\ \uparrow \end{bmatrix}_{\text{orb}}$$

* note : this is called "periforal coords" ↗

GROUND TRACK INTUITION



* Both $\omega = 0^\circ, 90^\circ$ & $e = 0, 0.3, 0.6$ swept

TIME frames of reference

UT1 : Solar time $\rightarrow$ [Y/M/D.ddd]

TAI : atomic time $\rightarrow [Y/M/D.ddd]$

UTC: leap-second adjusted $\rightarrow [Y/M/D.ddd]$

UT1 & TAI compr.

GMST: $\odot$ expressed as earth $\rightarrow$ [rad]
rot.

MJD: astrophysics friendly. $\rightarrow$ [float]

SPACE frames of reference

$$\underline{ECI \leftrightarrow ECEF} \quad \vec{r}_{ECEF} = R_z(90^\circ) \vec{r}_{ECI}$$

$\underline{ECEF} \leftrightarrow \underline{ENU}$ Need: ϕ (lat): $(-\frac{\pi}{2}, \frac{\pi}{2})$

or need $B \leftrightarrow \Phi$ $\lambda(\tan); (-\pi, \pi)$
 $L \leftrightarrow \lambda$ for geodesic

ALGORITHM: Cartesian ZOE

① find $\vec{n} = \frac{\vec{r}_1 \times \vec{r}_2}{|\vec{r}_1 \times \vec{r}_2|}$

② Find Ω , i , a , e directly

$$\Omega = \arctan(-\bar{h}_x / \bar{h}_y) \quad ; \quad i = \arctan(\sqrt{\bar{h}_x^2 + \bar{h}_y^2} / \bar{h}_z)$$

$$a = \left(\frac{2}{r} - \frac{\dot{r}^2}{\mu} \right)^{-1} ; \quad e = \sqrt{1 - \frac{p}{a}} = \left| \frac{(\vec{r} \times \dot{\vec{r}}) \times \dot{\vec{r}}}{\mu} - \frac{\vec{r}}{r} \right|$$

* Inverse vis-à-vis

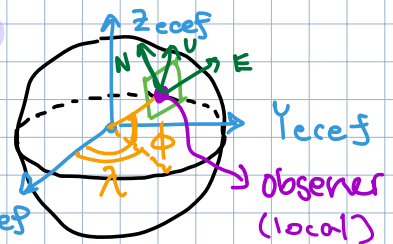
③ Find u (arg of latitude) & E (eccentric anom.) as useful intermediates.

$$u = \omega + \nu = \arctan(r_z / (r_x \overline{h}_y + r_y \overline{h}_x))$$

$$E = a \tan\left(\frac{\vec{r}_1 \cdot \vec{r}_2}{a^2 n}, 1 - \frac{r}{a}\right) \rightarrow \text{recall } n = \sqrt{\frac{\mu}{\epsilon_0}}$$

④ Find v then ω

$$\nu = \arctan(\sqrt{1-e^2} \sin E, \cos E - e) ; \quad \omega = u - \nu$$



GVEs → fundamental question is how to treat orbital perturbation term $\ddot{\vec{r}} = -\frac{\mu}{r^3} \vec{r} + \vec{f}$

the fundamental difeq: $\ddot{\vec{O}} = \ddot{\vec{O}}_{2b} + \frac{\partial \vec{g}}{\partial \vec{v}} \cdot \vec{f}$ where $\vec{O} = \vec{g}(\vec{r}, \vec{v}, t)$

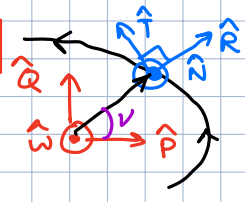
$\vec{f} = f_R \hat{R} + f_T \hat{T} + f_N \hat{N}$ → step (1) is to convert perturbing forces into RTN. Aside on RTN coords

Effects on OE (gen. & circ. cases)

* note: for circular case where ω, e are degenerate, modified terms used

$\vec{x}_{RTN} = R_S(\nu) \vec{x}_{PAW}$

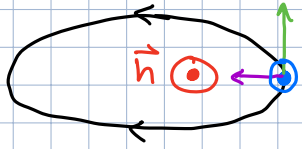
* $\hat{T}$ in RTN is NOT necessarily tangent to orbit!



	GENERAL in-plane vs out-plane	CIRC
$\frac{da}{dt}$	$\frac{2e \sin \nu}{n \sqrt{1-e^2}} f_R + \frac{2a \sqrt{1-e^2}}{nr} f_T$	$\frac{2}{n} f_T$
$\frac{de}{dt} \left(\frac{dex}{dt} \right)$	$\frac{\sqrt{1-e^2} \sin \nu}{na} f_R + \frac{\sqrt{1-e^2}}{na^2 e} \left(\frac{a^2(1-e^2)}{r} - r \right) f_T$	$\frac{1}{na} (\sin u f_R + 2 \cos u f_T)$
$\frac{di}{dt}$	$\frac{r \cos(\omega + \nu)}{na^2 \sqrt{1-e^2}} f_N$	$\frac{\cos u}{na} f_N$
$\frac{d\Omega}{dt}$	$\frac{r \sin(\omega + \nu)}{na^2 \sqrt{1-e^2} \sin i} f_N$	$\frac{\sin u}{na \sin i} f_N$
$\frac{d\omega}{dt} \left(\frac{dey}{dt} \right)$	$-\frac{\sqrt{1-e^2} \cos \nu}{nae} f_R + \frac{\sqrt{1-e^2}}{nae} \frac{2 + e \cos \nu}{1 + e \cos \nu} \sin \nu f_T - \frac{r \cot i \sin(\omega + \nu)}{na^2 \sqrt{1-e^2}} f_N$	$\frac{\sin u}{na} f_N$
$\frac{dM}{dt} \left(\frac{du}{dt} \right)$	$n - \frac{1}{na} \left(\frac{2r}{a} - \frac{1-e^2}{e} \cos \nu \right) f_R - \frac{1-e^2}{nae} \left(1 + \frac{r}{a(1-e^2)} \right) \sin \nu f_T$	$n + \frac{1}{na} (-2f_R - \sin u \cot i f_N)$

Types of orbital perturbations

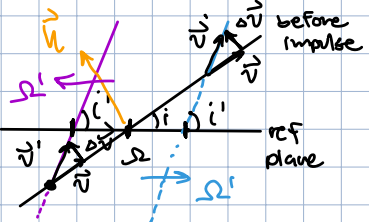
- In-plane parallel
- In-plane perpendicular
- Out of plane
- * pictured in (+) convention



ORBITAL PERTURBATIONS (Intuitions on directions of change)

out-of-plane ($\pm \vec{h}$)

a - no change ; e - no change ; intuition: changes only orbital plane

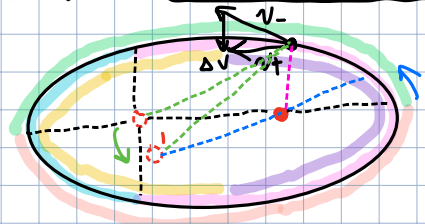


- i - Ascending ($-\frac{\pi}{2} < u < \frac{\pi}{2}$) $\Rightarrow i \uparrow$; descending ($\frac{\pi}{2} < u < \frac{3\pi}{2}$) $\Rightarrow i \downarrow$
- Ω - northern hemisphere ($0 < u < \pi$) $\Rightarrow \Omega \uparrow$ (nodal adv)
- southern hemisphere ($\pi < u < 2\pi$) $\Rightarrow \Omega \downarrow$ (nodal regr.)
- ω - opposite of Ω

In-plane - perpendicular (inward)

a - no change ; Ω, i - no change

intuition: changes orbital shape



- * Tip: imagine for $0 < \nu < \pi \Rightarrow e \downarrow$; for $\pi < \nu < 2\pi \Rightarrow e \uparrow$ (foci get closer)
- focus progresses for near apocenter $\Rightarrow \omega \uparrow$; near pericenter $\Rightarrow \omega \downarrow$ (advanced ω)
- (foci get farther)
- (regressed ω)

In-plane - Parallel (speed up)

Ω, i - no change ; intuition: changes orbital energy

a - increases by vis-visa
 $a = \left(\frac{2}{r} - \frac{v^2}{\mu} \right)^{-1} \Rightarrow v \uparrow, a \uparrow$

- e - to the left of minor axis - $e \downarrow$; to the right of minor axis - $e \uparrow$
- along minor axis - $e \rightarrow$
- ω - for ($0 < \nu < \pi$) $\omega \uparrow$ (adv)
- for ($\pi < \nu < 2\pi$) $\omega \downarrow$ (regresses)

J2 EFFECTS

perturbations in the RTN plane

$f_R = -\frac{3\mu J_2 R_E^2}{2r^4} (1 - 3 \sin^2 i \sin^2 u)$
 $f_T = -\frac{3\mu J_2 R_E^2}{2r^4} (\sin^2 i \sin u \cos u)$
 $f_N = -\frac{3\mu J_2 R_E^2}{2r^4} (\sin i \cos i \sin u)$

→ GVEs + avg across orbit

secular drift

$\dot{a}_{sec} = \dot{e}_{sec} = \dot{i}_{sec} = 0$
 $\dot{\omega}_{sec} = \frac{3}{2} n J_2 \left(\frac{R_E}{p} \right)^2 \left(2 - \frac{5}{2} \sin^2 i \right)$
 $\dot{\Omega}_{sec} = -\frac{3}{2} n J_2 \left(\frac{R_E}{p} \right)^2 \cos i$
 $\dot{M}_{sec} = n + \frac{3}{4} n J_2 \left(\frac{R_E}{p} \right)^2 \sqrt{1-e^2} (3 \cos^2 i - 1)$

- a has small stochastic periodic drift
- e has one periodic variation per orbit with max @ pericenter
- i has 2 periodic vars per orbit with max @ nodes.

$J_2 = 1.08263 \times 10^{-3}$ [dimensionless]

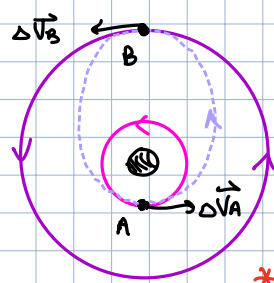
* since in RTN, ready to feed through GVEs

$\dot{\omega}$: $\dot{\Omega}$: MONOTONIC CHANGE

ORBITAL TRANSFERS

Prelim: Oberth effect: burn at perigee for $\uparrow V$, apogee for $\downarrow V$

Hohmann transfer (circular $\rightarrow$ circular)
(ALL BURNS TANGENTIAL $\phi_{\pm A} = \phi_{\pm B} = 0$)



$$a_t = \frac{r_A + r_B}{2}$$

$$V_A = \sqrt{\mu/r_A}$$

$$V_B = \sqrt{\mu/r_B}$$

$$V(r) = \sqrt{2\mu \left(\frac{1}{r} - \frac{1}{r_A + r_B} \right)}$$

* velocity along ANY point on the transfer orbit

$$\Delta V_{A/B} = |\vec{V}_{A/B} - \vec{V}_{-A/B}|$$

* ΔV always positive

$$t_{trans} = \pi \sqrt{\frac{(r_A + r_B)^3}{8\mu}}$$

Notes on Hohmann:

① Can be repurposed for coaxial ellipse orbital transfer

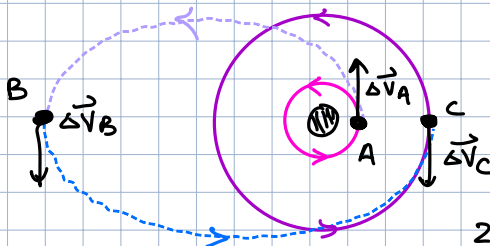
② Optimizes for min ΔV when $\frac{r_f}{r_i} < 11.94$

Little Oberth proof: $\Delta \mathcal{E} = \mathcal{E}_+ - \mathcal{E}_-$

$$= \frac{(\vec{V} + \Delta \vec{V}) \cdot (\vec{V} + \Delta \vec{V})}{2} - \frac{\mu}{r} - \left(\frac{\vec{V} \cdot \vec{V}}{2} - \frac{\mu}{r} \right) = \vec{V} \cdot \Delta \vec{V} + \frac{\Delta \vec{V} \cdot \Delta \vec{V}}{2}$$

Key term implies $\Delta \vec{V}$ best applied $\parallel \vec{V}$ & when $\vec{V}$ is maxed

Bi-elliptic transfer (alternative to Hohmann)



$\Delta \vec{V}_A$ & $\Delta \vec{V}_B$ are speed-up maneuvers

$\Delta \vec{V}_C$ is a slow-down maneuver.

2-ellipses! one, two

Notes: ① more efficient than Hohmann for $r_f/r_i > 11.94$

② Can use all the same relations as Hohmann.

③ Key design decision: $r_{intermediate}$ (is a non-convex optimization problem).

NON-COPLANAR MANEUVERS

Prelim: plane changes are expensive in ΔV , so apply them when $\vec{V}$ is SMALLEST! (inverse Oberth)

Change i ONLY: apply $\Delta \vec{V}$ @ ascending or descending node & apply

$$\Delta V = 2V \sin \frac{\Delta i}{2}$$

* isosceles triangle identity

Change Ω ONLY: is generally impossible unless orbit is polar, then apply $\Delta \vec{V}$ at poles.

Combined maneuvers (perigee & apogee kicks)

$\rightarrow$ design decision is Δi_A vs Δi_B to achieve desired Δi_{tot}

process: ① based on initial & final orbits, define $V_{A+/-}$, $V_{B+/-}$ as in Hohmann.

② Solve the following optimization problem:

$$\min (\Delta V_A + \Delta V_B) \text{ s.t. } \Delta i_A + \Delta i_B = \Delta i_{tot}$$

Δi_A

Law of cosines.

$$\Delta V_A = \sqrt{V_{A-}^2 + V_{A+}^2 - 2V_{A-}V_{A+} \cos \Delta i_A}$$

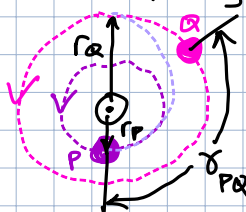
$$\Delta V_B = \sqrt{V_{B-}^2 + V_{B+}^2 - 2V_{B-}V_{B+} \cos \Delta i_B}$$

ORBITAL PHASING

Planet Q (n_Q mean motion)

Planet P (n_P mean motion)

Hohmann phasing: timing of Hohmann transfer



$$\delta_{PQ} = \pi - n_Q t_{trans}$$

$$\text{Recall: } n_Q = \sqrt{\mu/r_Q^3}$$

$$t_{trans} = \pi \sqrt{\frac{(r_P + r_Q)^3}{8\mu}}$$

Synodic period

$$T_{syn} = 2\pi / |n_P - n_Q|$$

* how often ideal δ_{PQ} occurs!

* notice the closer n_P & n_Q , the less frequent δ_{PQ} !

THE ROCKET EQN

$$F_{sc} = \dot{m}_{sc} a; F_{sc} = -V_{exit} \dot{m}_{sc}$$

$$\int \frac{F_{sc}}{\dot{m}_{sc}} dt = \int a dt \Rightarrow \int \frac{V_{exit} \dot{m}_{sc}}{\dot{m}_{sc}} dt = \int \frac{dv}{dt} dt$$

$$\Rightarrow -V_{exit} \int_{m_{sc}}^{m_{sc} - \Delta m} \frac{1}{m_{sc}} dm_{sc} = \int_v^{v+\Delta v} dv$$

$$\Rightarrow \Delta V = V_{exit} \ln \left(\frac{m_{sc}}{m_{sc} - \Delta m_P} \right)$$

* The ROCKET EQN:

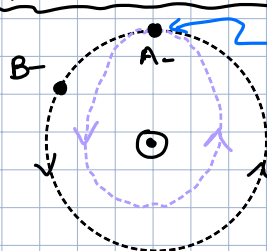
* notice ΔV is only dependent on Δm_P , not m_P

For small $\Delta m_P \ll m_{sc}$

apply $\ln(1-x) \approx -x$ for $x \rightarrow 0$

$$\Delta V \approx \frac{V_{exit}}{m_{sc}} \Delta m_P$$

PHASING WITHIN CIRCULAR ORBITS



INTERCEPT

(ΔV lost; A "catches up" to B)

PENDEZ - VOUS

(ΔV applied; A "falls behind")

ORBITAL TRANSFERS AUX.

Hohmann

ΔV_{tot} is maxed at $r_2/r_1 = 15.58$

$$V_{min} = \sqrt{\frac{\mu}{a} \left(\frac{1-e}{1+e} \right)}$$

$$V_{max} = \sqrt{\frac{\mu}{a} \left(\frac{1+e}{1-e} \right)}$$

Bi-elliptic

$$\Delta V_A = \sqrt{\frac{2\mu}{r_A} - \frac{\mu}{a_1}} - \sqrt{\frac{\mu}{r_A}}$$

$$\Delta V_B = \sqrt{\frac{2\mu}{r_B} - \frac{\mu}{a_2}} - \sqrt{\frac{\mu}{r_B}}$$

$$\Delta V_C = \sqrt{\frac{2\mu}{r_C} - \frac{\mu}{a_2}} - \sqrt{\frac{\mu}{r_C}}$$

COMMON ORBITS

Simple orbits

VLEO: $e \sim 0$, $h = 100 - 400$ km
 LEO: $e \sim 0$, $h = 400 - 1200$ km
 MED: $e \sim 0$, $2000 - 35,786$ km
 GEO: $e = 0$, $h = 35,786$ km
 $i = 0$, the same
 GSO: e is free, $a = 42,164$ km
 i is free
 Graveyard: GEO + 300 km
 GPS: $e \sim 0$; $a \sim 20,000$ km

FROZEN ORBITS: $i \mid \dot{\omega}_{sec} = 0$

- $\dot{\omega}_{per}$ has higher amplitude as $r \downarrow$
- Why? $\rightarrow J_2$ effects scale by $\frac{1}{r^4}$

Molniya (frozen orbit)

$a = 26,600$ km; $e = 0.74$ * highly eccentric
 $i = 63.4^\circ$; $T = \frac{1}{2}$ sidereal day
 * frozen!

Tundra (frozen + GSO)

$a = 42,164$ km; $e = 0.2 - 0.4$
 * GSO
 $T \sim 1$ sidereal day; $i = 63.4^\circ$
 * frozen!

QZSS (JAXA)

$a = 42,164$ km; $e \sim 0.075$
 * GSO
 $i \sim 43^\circ$; $T \sim 1$ sidereal day

MOLNIYA ($\omega = 270^\circ$)



TUNDRA ($\omega = 270^\circ$)

QZSS

SSO ($\dot{\Omega} = 360^\circ/\text{year} = 0.9856^\circ/\text{day}$)

$i = 97 - 99^\circ$ * slightly retrograde! $\rightarrow$ exact design depends on a & e in P

$T \sim 90 - 120$ minutes;

Dusk-Dawn orbit (an extension of SSO where the ascending node is always at 6PM/AM local).

ONE-STOP

$$\ddot{\vec{r}} + \frac{\mu}{r^3} \vec{r} + \vec{f} = 0 \quad \text{* General 2-body}$$

$$\mathcal{E} = \frac{E}{m} = \frac{1}{2} v^2 - \frac{\mu}{r} \quad \text{* specific orbital energy}$$

Circular Orbits

$$v_{circ}(r) = \sqrt{\frac{\mu_{ce}}{r}} \quad T_{circ} = 2\pi \sqrt{\frac{r^3}{\mu}}$$

The road to OES

$$\vec{h} = \vec{L}/m = \vec{r} \times \dot{\vec{r}} \quad ; \quad \frac{d\vec{h}}{dt} = 0$$

$$\vec{A} = \dot{\vec{r}} \times \vec{h} - \mu \frac{\vec{r}}{r} \quad \text{* eccentricity vector}$$

$$e = \frac{|\vec{A}|}{\mu} \quad p = \frac{h^2}{\mu}$$

$$v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right) \Rightarrow a = \left(\frac{2}{r} - \frac{v^2}{\mu} \right)^{-1}$$

* vis-viva relation & inverse

$$\Delta A = \frac{1}{2} |\dot{\vec{r}} \times \ddot{\vec{r}} \Delta T| = \frac{1}{2} \Delta T |\dot{\vec{h}}| \quad \text{* Kepler's 2nd}$$

$$\frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3}; T^2 = 4\pi^2 \frac{a^3}{\mu} \quad \text{* Kepler's 3rd}$$

$$\begin{bmatrix} \vec{r} \\ \dot{\vec{r}} \end{bmatrix}_{ECI} = R_3(-\Omega) R_1(-i) R_3(-\omega) \begin{bmatrix} \vec{r} \\ \dot{\vec{r}} \end{bmatrix}_{PQW}$$

* Perifocal to ECI & vice versa * Rocket equations

$$\vec{r}_{ECIF} = R_3(\Theta) \vec{r}_{ECI}$$

* ECEF $\leftrightarrow$ ECI conversion

* ECEF $\rightarrow$ ENV

$$\begin{bmatrix} E \\ N \\ V \end{bmatrix} = R_1\left(\frac{\pi}{2} - \phi\right) R_3\left(\frac{\pi}{2} + \lambda\right) \begin{bmatrix} x - x_{local} \\ y - y_{local} \\ z - z_{local} \end{bmatrix}_{ECIF} \quad \text{observer dependent}$$

$$\phi = \text{atan}\left(\frac{z_{local}}{\sqrt{x_{local}^2 + y_{local}^2}}\right) \quad \lambda = \text{atan}\left(\frac{y_{local}}{x_{local}}\right)$$

$$\vec{x}_{RTN} = R_3(V) \vec{x}_{PQW}$$

* perifocal to RTN

$$v(r) = \sqrt{2\mu \left(\frac{1}{r} - \frac{1}{r_A + r_B} \right)}$$

* velocity anywhere along Hohmann transfer

$$t_{trans} = \pi \sqrt{\frac{(r_A + r_B)^3}{8\mu}}$$

* Hohmann transfer time

$$F_{sc} = -V_{exit} \dot{m}_{sc}$$

$$\Delta V = V_{exit} \ln\left(\frac{m_{sc}}{m_{sc} - \Delta m_p}\right)$$

$$\Delta V \approx \frac{V_{exit}}{m_{sc}} \Delta m_p$$

* Rocket equations

$$\dot{\omega}_{sec} = \frac{3}{2} n J_2 \left(\frac{R_E}{p} \right)^2 \left(2 - \frac{5}{2} \sin^2 i \right)$$

$= 0$ @ $i = 63.4^\circ$ & 116.6°

$$\dot{\Omega}_{sec} = -\frac{3}{2} n J_2 \left(\frac{R_E}{p} \right)^2 \cos i$$

$= 1 \text{ rev/year}$ @ $i \sim 97 - 99^\circ$

* J_2 key secular perturbations

RELEVANT MATH:

Polar coords (r, θ, z)

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \text{atan}(y/x) \quad [0, 2\pi] \text{ (azm.)}$$

Spherical coords (r, ϕ, λ)

$$r = \sqrt{x^2 + y^2 + z^2}$$

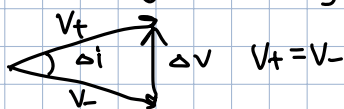
$$\lambda = \text{atan}(y/x) \text{ (lon.)}$$

$$\phi = \text{atan}\left(\frac{z}{\sqrt{x^2 + y^2}}\right) \text{ (lat.)}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = r \begin{bmatrix} \cos \phi \cos \lambda \\ \cos \phi \sin \lambda \\ \sin \phi \end{bmatrix}$$

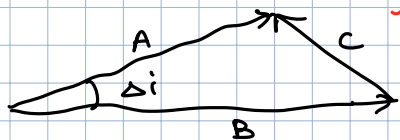
useful triangle properties

Isosceles triangle property



$$\Delta V = 2V \sin \frac{\Delta i}{2}$$

Law of cosines * works for ANY triangle



$$C = \sqrt{A^2 + B^2 - 2AB \cos \Delta i}$$

Deep-dive on spherical harmonics

Solution to Laplace's equation $\nabla^2 U(\vec{r}) = 0$ is the decomp $U(r, \lambda, \phi) = Q_1(r) Q_2(\lambda) Q_3(\phi)$

Fully expanded spherical harmonic series:

$$U(r, \lambda, \phi) = -\frac{GM}{r} \left[1 + \sum_{n=2}^{\infty} \sum_{m=0}^n \left(\frac{R_0}{r}\right)^n (C_{nm} \cos(m\lambda) + S_{nm} \sin(m\lambda)) P_{nm} \sin \phi \right]$$

Separated into zonal & sectoral/tesseral terms

$$U(r, \lambda, \phi) = -\frac{GM}{r} \left[1 - \left(\frac{R_0}{r}\right)^n J_n P_n \sin \phi + \sum_{n=2}^{\infty} \sum_{m=1}^n \left(\frac{R_0}{r}\right)^n (C_{nm} \cos(m\lambda) + S_{nm} \sin(m\lambda)) P_{nm} \sin \phi \right]$$

* notice that zonal-term is only latitude (ϕ) dependent & $J_n = -C_{n0}$

where order $m = 0$.

Intuition

'Zonal' terms : $m=0$

'Sectoral' terms : $m=n$

$(n-m)$: # of zero-crossings in latitude

$2m$: # of zero crossings in longitude

for $m=0$

$n=0$: assumes spherical uniformity J2!

$n=1$: captures COM-off-center inaccuracies

$n=2$: captures earth-oblateness

$n=3+$: captures other terms

Pure n : earth is cut ($m=0$) like a potato

sectoral ($m=n$) : earth is cut like an orange

tesseral : earth is checkerboard.

useful trig identities

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

Rotation matrices

* The convention here is a rotation of the coordinate system itself

* For rotation of a point in a fixed coordinate system, use transposes

$$R_1(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}$$

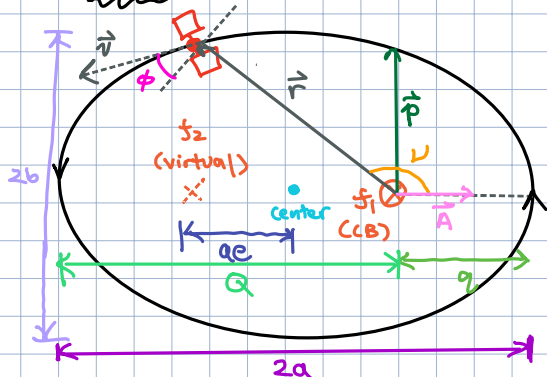
$$R_2(\theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_3(\theta) = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

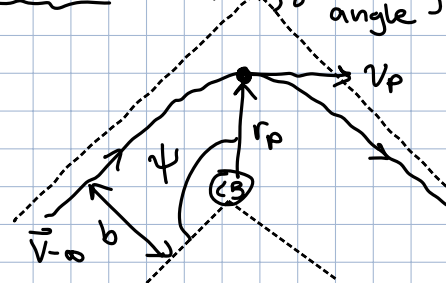
Conic sections

$$r = \frac{p}{1 + e \cos(\nu)} \quad \text{+ General form.}$$

Ellipse: $0 < e < 1$



Hyperbola: $e > 1$



$$Q = r(\nu = \pi) = \frac{p}{1-e} ; q = r(\nu = 0) = \frac{p}{1+e}$$

$$a = \frac{1}{2}(Q+q) = \frac{p}{1-e^2} ; p = a(1-e^2)$$

$$r = \frac{a(1-e^2)}{1+e \cos(\nu)} ; e = \frac{Q-q}{Q+q} = \sqrt{1-b^2/a^2}$$

$$b = a\sqrt{1-e^2} ; \nu_{@b} = \cos^{-1}(e)$$

$$\text{Area} = \pi ab$$

$$V_{\infty}^2 = V^2 - V_{esc}^2 \text{ (hyperbolic excess speed)}$$

$$\sin \frac{\delta}{2} = \frac{1}{e} \Rightarrow \delta = 2 \arcsin\left(\frac{1}{e}\right)$$

$$\psi = \arccos(-1/e)$$

$$\epsilon = \frac{V_{\infty}^2}{2} = \frac{V_p^2}{2} - \frac{\mu}{r_p}$$

$$e = \sqrt{1 + \frac{2\epsilon h^2}{\mu^2}} = \sqrt{1 + \frac{b^2 V_{\infty}^4}{\mu^2}} = 1 + \frac{r_p V_{\infty}^2}{\mu}$$

$$b = \frac{\mu}{V_{\infty}^2} \sqrt{e^2 - 1}$$